

DAY 2 – Conditions & Loops Lab Sheet

SECTION A: Conditional Statements

Question 1: Write a program to accept a number **N** and print whether it is **Positive, Negative, or Zero**.

Fundamentals:

- if, else if, else
- Comparison with 0

Logic:

- $> 0 \rightarrow$ Positive
- $< 0 \rightarrow$ Negative
- $== 0 \rightarrow$ Zero

Code:

```
import java.util.Scanner;

public class CheckNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        if (n > 0)
            System.out.println("Positive");
        else if (n < 0)
            System.out.println("Negative");
        else
            System.out.println("Zero");
    }
}
```

Question 2: Write a program to accept **two numbers** and print the **greater value**.

Fundamentals:

- if-else
- > operator

Logic:

- Compare a and b

Code:

```
import java.util.Scanner;

public class GreaterNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();

        if (a > b)
            System.out.println(a);
        else
            System.out.println(b);
    }
}
```

Question 3: Write a program to check whether a number is **EVEN** or **ODD**.

Fundamentals:

- Modulus %

Logic:

- $n \% 2 == 0 \rightarrow \text{Even}$
- $\text{else} \rightarrow \text{Odd}$

Code:

```
import java.util.Scanner;

public class EvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();

        if (n % 2 == 0)
            System.out.println("Even");
        else
            System.out.println("Odd");
    }
}
```

Question 4: Write a program to accept **two numbers** and print whether their **sum** is **EVEN** or **ODD**.

Fundamentals:

- + operator
- %

Logic:

- $sum = a + b$
- Check $sum \% 2$

Code:

```
import java.util.Scanner;

public class SumEvenOdd {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int a = sc.nextInt();
        int b = sc.nextInt();

        int sum = a + b;

        if (sum % 2 == 0)
            System.out.println("Even");
        else
            System.out.println("Odd");
    }
}
```

SECTION B: Looping Constructs

Question 5: Write a program to print numbers from 1 to 100.

Fundamentals:

- for loop

Logic:

- Start = 1, End = 100, Step = +1

Code:

```
public class Print1To100 {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 100; i++) {  
            System.out.print(i + " ");  
        }  
    }  
}
```

Question 6: Write a program to print **alternate numbers from 1 to 99**
(1, 3, 5, 7, ... 99)

Fundamentals:

- Increment by 2

Logic:

- $i = i + 2$

Code:

```
public class AlternateOdd {  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 99; i += 2) {  
            System.out.print(i + " ");  
        }  
    }  
}
```

Question 7: Write a program to print **alternate numbers from 0 to 100**
(0, 2, 4, 6, ... 100)

Fundamentals:

- Even numbers

Logic:

- Start from 0, increment by 2

Code:

```
public class AlternateEven {  
    public static void main(String[] args) {  
  
        for (int i = 0; i <= 100; i += 2) {  
            System.out.print(i + " ");  
        }  
    }  
}
```

Question 8: Write a program to print numbers from **100 to 0**.

Fundamentals:

- Reverse loop

Logic:

- Start = 100, Decrement

Code:

```
public class Reverse100To0 {  
    public static void main(String[] args) {  
  
        for (int i = 100; i >= 0; i--) {  
            System.out.print(i + " ");  
        }  
    }  
}
```

Question 9: Write a program to print numbers from **100 to 1 by skipping 2 numbers** (100, 97, 94, ...)

Fundamentals:

- Decrement by 3

Logic:

- $i = i - 3$

Code:

```
public class ReverseSkip {  
    public static void main(String[] args) {  
  
        for (int i = 100; i >= 1; i -= 3) {  
            System.out.print(i + " ");  
        }  
    }  
}
```